

Introduction to Environmental Science EES102

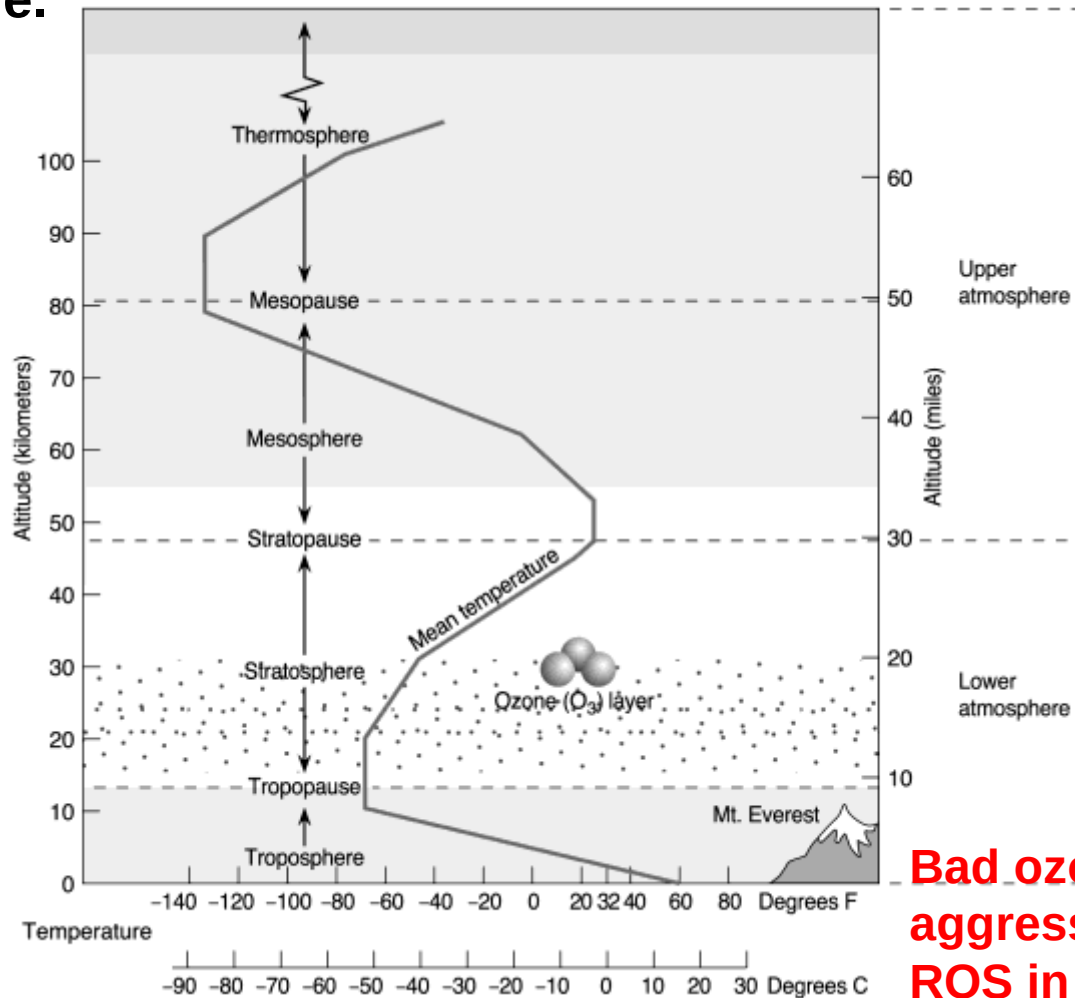
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OUTDOOR AIR POLLUTION

NATIONAL AMBIENT AIR QUALITY STANDARD

Pollutant	Time weighted average	Concentration limit
Sulphur dioxide (SO ₂) [µg m ⁻³]	Annual & 24 hrs	50 & 80
Nitrogen oxide (NO) + Nitrogen dioxide (NO ₂) = NO _x [µg m ⁻³]	Annual & 24 hrs	40 & 80
Ozone (O ₃) [µg m ⁻³]	8 hrs & 1 hr	100 & 180
Carbon monoxide (CO) [mg m ⁻³]	8 hrs & 1 hr	2 & 4
Ammonia (NH ₃) [µg m ⁻³]	Annual & 24 hrs	100 & 400
Benzene (C ₆ H ₆) [µg m ⁻³]	Annual	5
PM _{2.5} [µg m ⁻³]	Annual & 24 hrs	40 & 60
PM ₁₀ [µg m ⁻³]	Annual & 24 hrs	60 & 100
Arsenic (As) [ng m ⁻³]	Annual	6
Nickel (Ni) [ng m ⁻³]	Annual	20
Benzo(α)Pyrene (BAP) in aerosol [ng m ⁻³]	Annual	1

The earth's atmosphere is subdivided vertically into four major regions based on the air-temperature profile. The ozone layer that protects us from the sun's UV radiation is in the stratosphere.



Good ozone

Bad ozone =
aggressive oxidant =
ROS in lung

Never buy an air purifier that promises to oxidize pollutants or humidify the air. (No UV lamp no humidifier). Hepa filters and activated charcoal are great though

MICROGRAMS PER CUBIC METER & PPB

We usually measure the concentration of air pollutants ppb because in this unit measurements are pressure and temperature independent

1 ppb of x means one molecule of x in a billion of molecules of air

The air quality standard is in microgram per m³ which means we have to convert what we measure and the conversion is pressure and temperature dependent

MICROGRAMS PER CUBIC METER & PPB

For an atmospheric concentration of NO₂ that is 50 ppb

$$\text{ppb NO}_2 = \frac{\text{molecules of NO}_2}{1 \text{ billion molecules of air}}$$

$$\text{ppb NO}_2 = \frac{\text{mol of NO}_2}{1 \text{ billion mol of air}}$$

If we want to convert that to microgram per meter cube, first we have to compute how many gram NO₂ there are in 50 ppb NO₂

Remember 12 g of ¹²C = 1 mol

$$46 \text{ g NO}_2 = \underline{1\text{-mol NO}_2}$$

MICROGRAMS PER CUBIC METER & PPB

Next, using the ideal gas law, we can compute how many liters of air one mol of air occupies

$$V = nRT/p = \frac{(1 \text{ mol}) (0.082 \text{ L atm mol}^{-1}\text{K}^{-1}) (T^\circ\text{K})}{(p \text{ atm})}$$

$$= 24.6 \text{ L @ 300 K and 1 atm}$$

$$(\text{g m}^{-3}) = \frac{\text{Molecular weight (g)} \times \text{concentration (ppb)} \times 1000 (\text{L /m}^{-3})}{\text{Volume (L)}}$$

$$\text{So } 46 \times 50 \times 10^{-9} [\text{g/mol} \cdot \text{mol}] / 0.0246 [\text{m}^3] = 93.5 \times 10^{-6} \text{ g m}^{-3} = 93.5 \mu\text{g m}^{-3}$$

If you don't worry about units just remember:

$$\frac{\text{Molecular weight (g)} \times \text{concentration (ppb)}}{\text{Volume (L)}} = \mu\text{g m}^{-3}$$

CO

Not very toxic at levels usually present

Interferes with the oxygen carrying capacity of blood.

Normally haemoglobin in red blood cells combines with oxygen in the lungs to form oxyhemoglobin which is carried in the blood stream

CO binds more strongly to haemoglobin than oxygen. If too much CO is present in the lungs it displaces oxygen from haemoglobin reducing the amount of oxygen the blood can carry

The gas is colorless, tasteless and odourless which is why people fall unconscious without realizing it is present. Symptoms are dizziness, drowsiness and impaired judgement.

Heating stoves, gas stoves and gas geysers are the main cause of death when used in unventilated spaces. CO is also the main cause of death in house fires.

LOS ANGELES SMOG (OZONE SMOG)

Observed in LA in the 1940s

- Hot days
- Strong sunshine

Strongly oxidating, causes ROS, lung, skin, eye-irritations

Plant damage

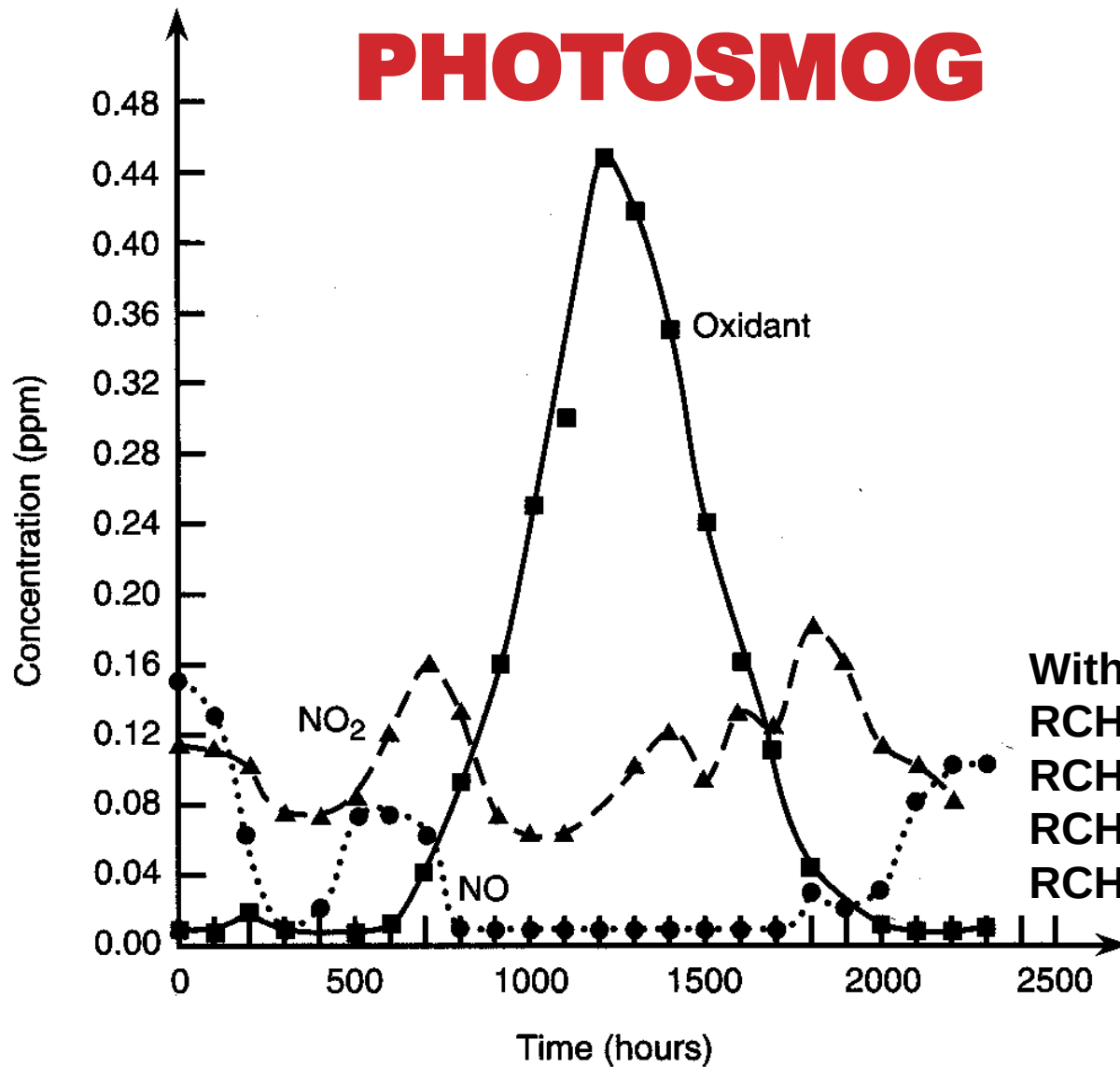
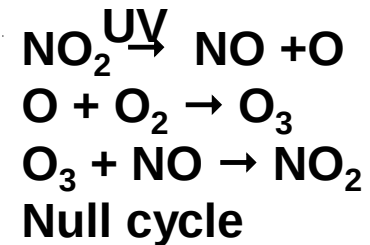
- Arie Haagen Smit: Plant damage reproduced in exp. with
 - Organic trace gases, NO_x , sunlight
 - Car exhaust, NO_x , sunlight

$\text{VOC} + \text{NO}_x + \text{sunlight} \rightarrow \text{O}_3 + \text{"other products"}$

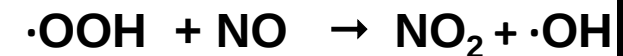
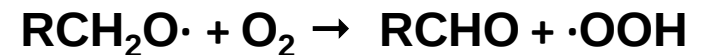
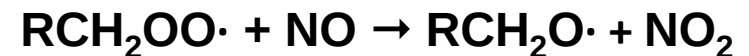
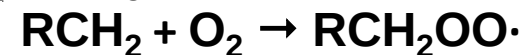
Today worldwide problem

PHOTOSMOG

High temperature combustion breaks N_2 bond and creates NO NO is oxidized to NO_2 in the atmosphere



With VOCs



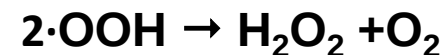
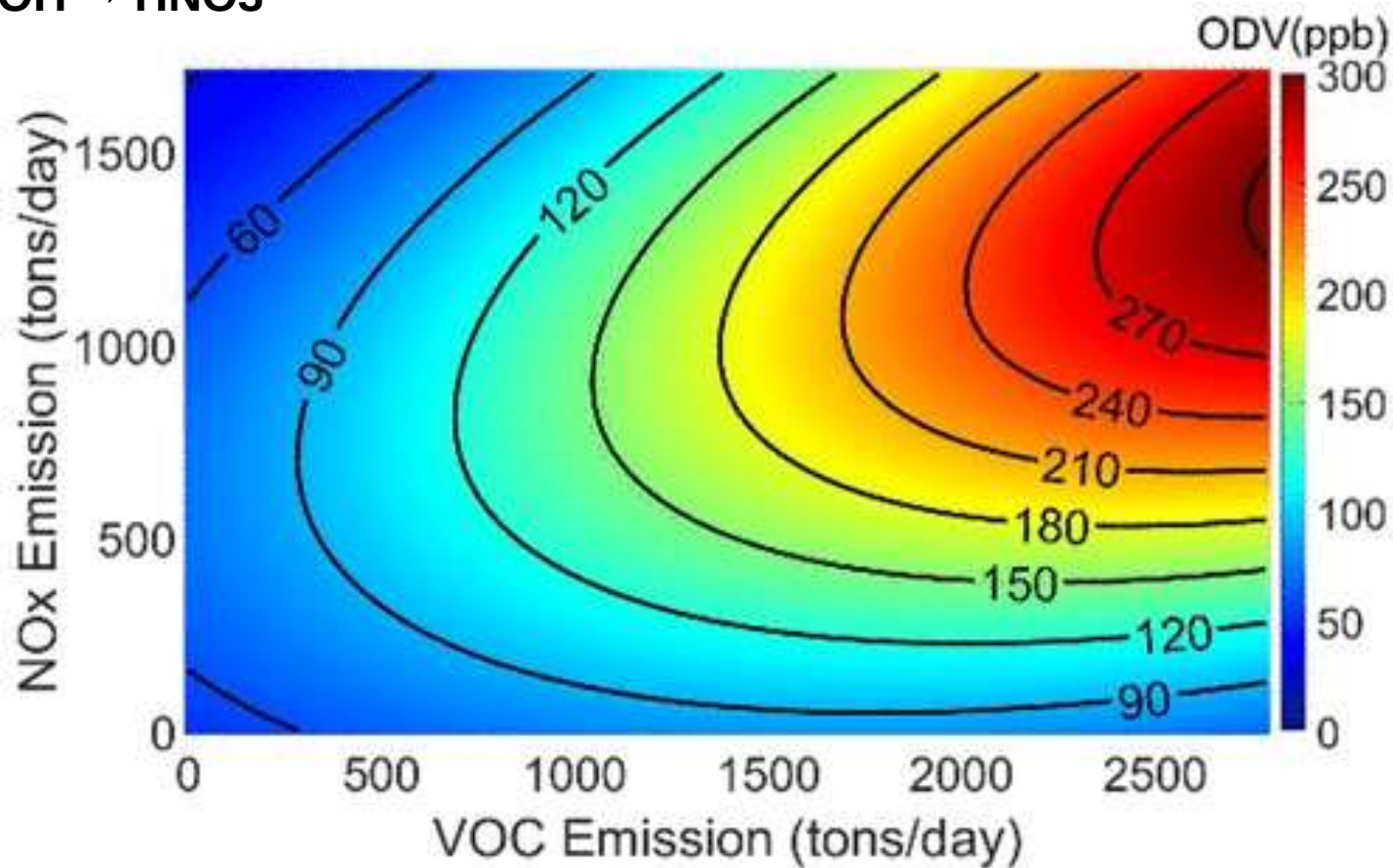
chain reaction

propagating radicals

every extra NO_2 gives you extra O_3

FIGURE 1.3 Diurnal variation of NO, NO_2 , and total oxidant in Pasadena, California, on July 25, 1973 (adapted from Finlayson-Pitts and Pitts, 1977).

PHOTOSMOG IS NON-LINEAR



LONDON SMOG

17. Jh., John Evelyn: report on air pollution in London from coal:

„It is this horrid Smoke which obscures our Church and makes our Palaces look old, which fouls our Cloth and corrupts the Waters, so as the very Rain, and refreshing Dews which fall in the several Seasons, precipitate to impure vapour, which with its black and tenacious quality, spots and contaminates whatever is exposed to it.“

„But, without the use of Calculation it is evident to every one who looks on the yearly Bill of Mortality, that near half the children that are born and bred in London die under two years of age^a. Some have attributed this amazing destruction to luxury and the abuse of Spirituous Liquors: These, no doubts, are powerful assistants; but the constant and unremitting Poison is communicated by the foul Air, which, as the Town still grows larger, has made regular and steady advances in its fatal influence“

^a „A child born in a Country Village has even the chance of living near 40 years...“

IMPORTANT REQUIREMENTS

- Inversions
- fog

Contaminants:

SO_2 , soot

$\Rightarrow \text{H}_2\text{SO}_4$, sulfate, aerosol

4000 people died
1952 in London

From SMOG...??

Predicted in a 1932 paper published after a similar episode in the Meuse Valley, Belgium, in December, 1930

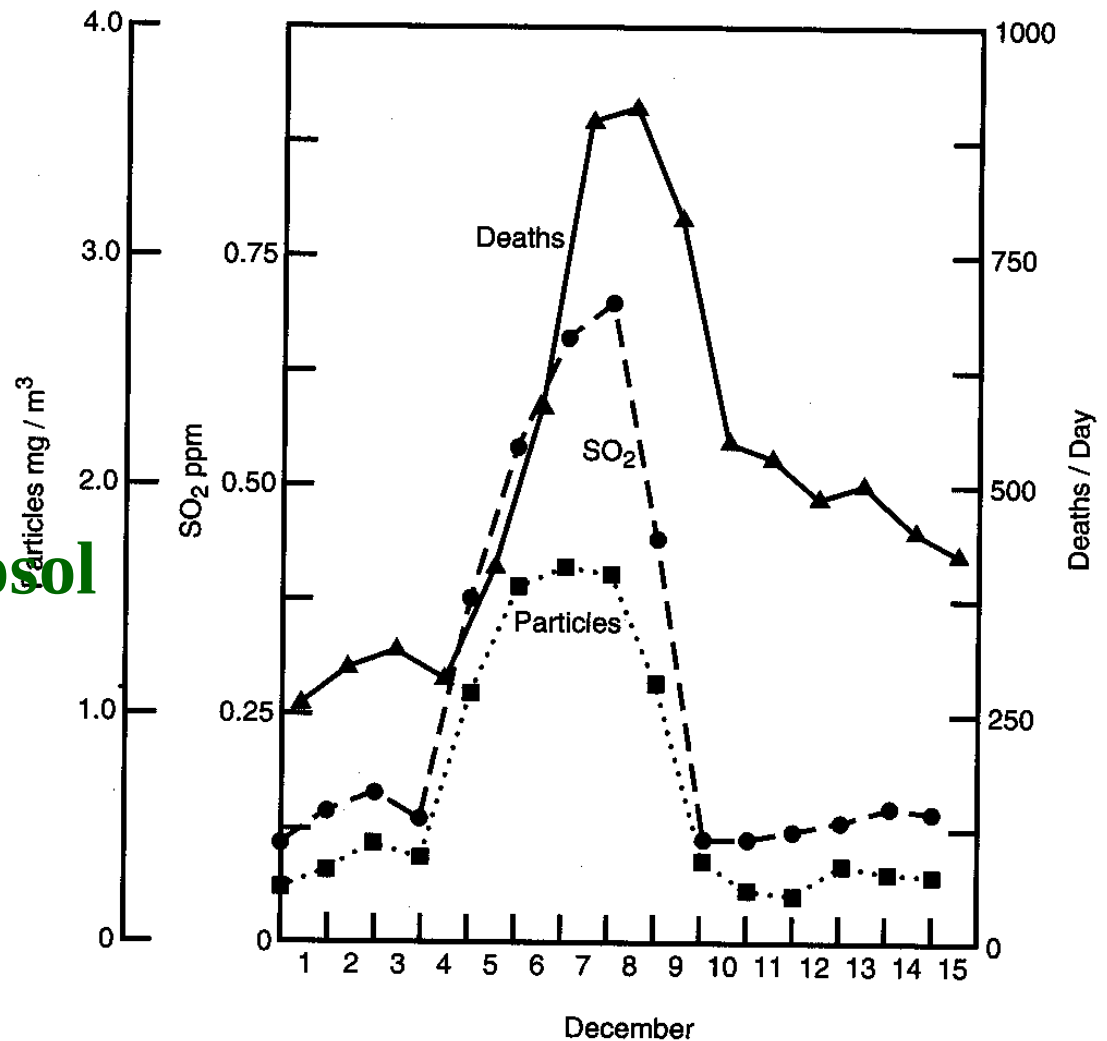
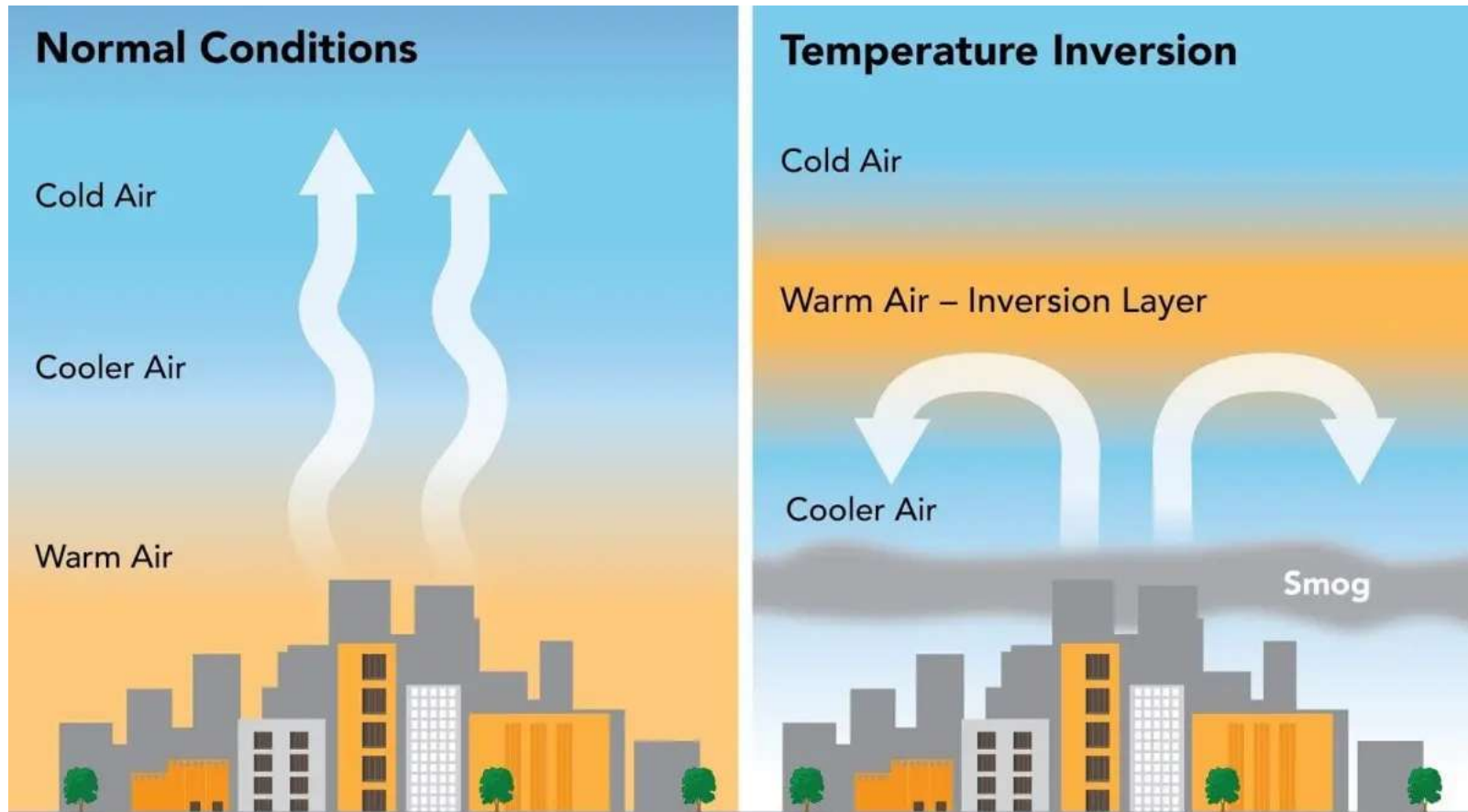


FIGURE 1.2 Concentrations of SO_2 and “smoke” as well as the death rate during the 1952 smog episode (adapted from Wilkins, 1954).

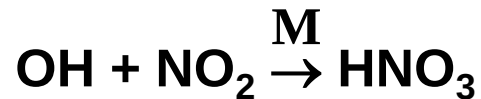
ATMOSPHERIC INVERSION



If the Bhopal accident had not happened during an inversion event the story might have ended differently

ACID RAIN, WATER POLLUTION

Term “Acid Rain”: Robert Angus Smith, 1872



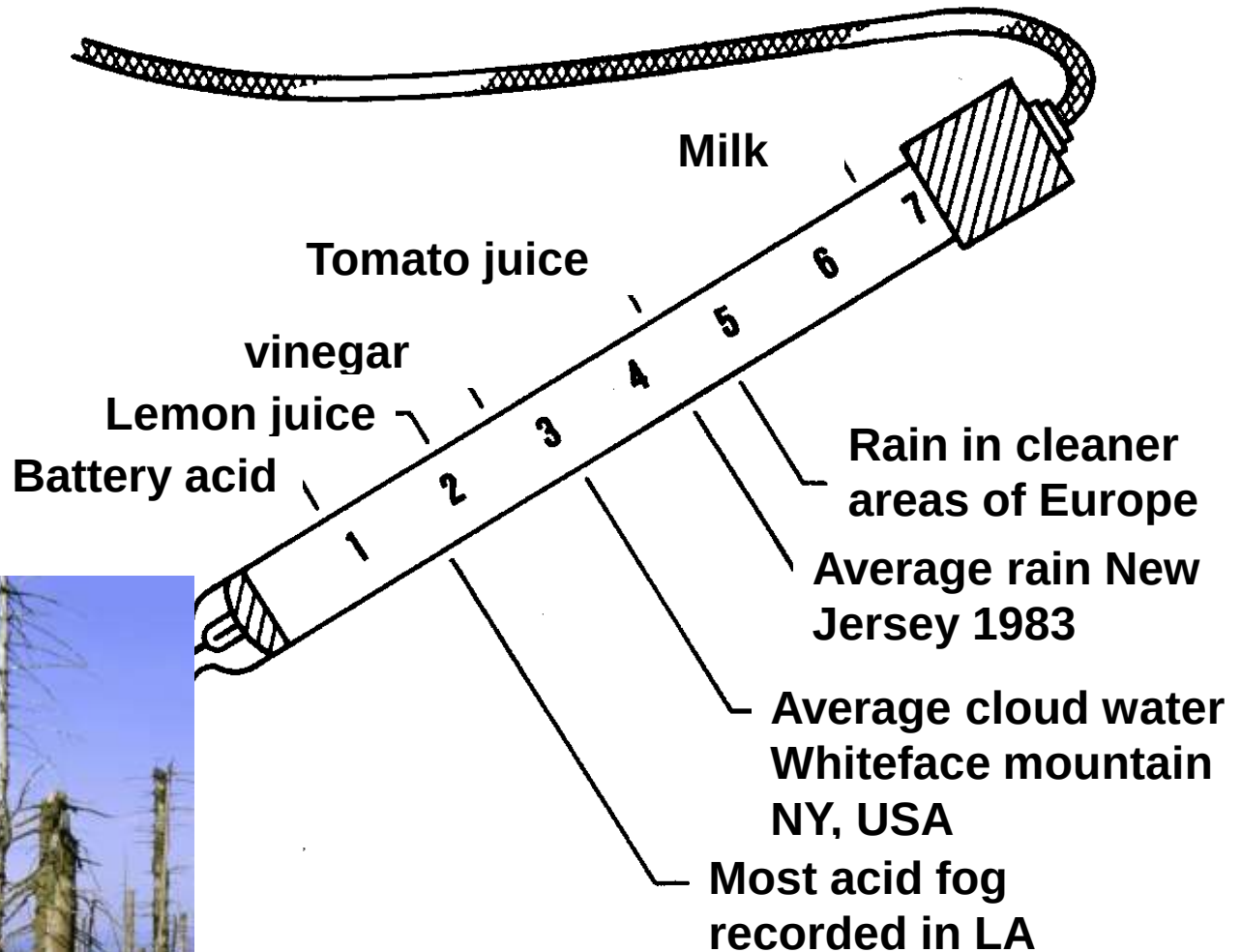
Acid rain: pH 4 close to cities

Fog: pH 1.69 meas. in California (Jacob and Hoffmann, 1983)

TALLER SMOKESTACKS



THE ACID RAIN PROBLEM



ATMOSPHERIC AEROSOL PARTICLES

PRIMARY AND SECONDARY AEROSOL PARTICLES

- Definition: Primary aerosol particles **are emitted directly as particles**
- Definition: Secondary aerosol particles **are formed in the atmosphere from gaseous emissions through gas to particle conversion**

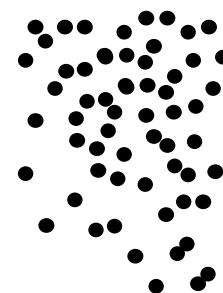


Natural primary aerosol
Mineral dust



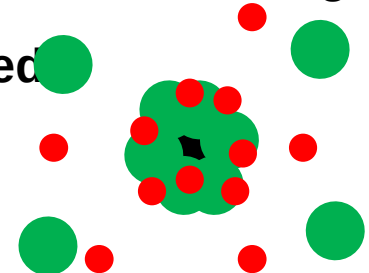
Anthropogenic
primary aerosol

Condensable vapour



freshly
nucleated
particle
nm

Condensation
and coagulation



ATMOSPHERIC AEROSOL PARTICLES

NATURAL VERSUS ANTHROPOGENIC SOURCES

- Definition: Aerosol particles from natural sources **are emitted without human intervention** (Preindustrial background aerosol burden)
- Definition: Anthropogenic aerosol particles **are man-made**



Sources	Approx. global emission flux (Tg a ⁻¹)
<i>Primary aerosols</i> Desert and semi-arid areas (mineral dust) Marine sea spray Volcanic dust Biogenic (pollens, debris) <i>Secondary aerosols</i> Sulphates from biogenic gases Sulphates from volcanic SO₂ Nitrates from nitrogen oxides Organics from volatile organic compounds (COV) TOTAL	1500 1300 33 50 90 12 55 22 ~3060
<i>Primary aerosols</i> Industrial dust Soot <i>Secondary aerosols</i> Organics from oxidation of volatile organic compounds Sulphates produced by oxidation of anthropogenic SO₂ Sulphates produced from biomass combustion gases Nitrates from nitrogen oxides TOTAL	100 10 10 190 90 50 ~390

If the natural burden is so high ($\sim 3060 \text{ Tg an}^{-1}$) and the anthropogenic contribution only slightly more than 13% of that (390 Tg an^{-1}), why do we attempt to reduce the aerosol burden? We can't do anything about the natural background, right?

➤ The chemical composition is different! Inhalation of sea salt aerosol is a treatment for many lung conditions (Chronic bronchitis, Asthma, ...) => Aerosol emitted by traffic can be their cause.

➤ The short lifetime of aerosol means the aerosol burden and its effects are highly local. Anthropogenic emissions are where we are!

Silicosis: Occupational lung disease

Silicosis is an often fatal lung disease caused by breathing dust containing crystalline silica particles, a basic component of sand and granite. There is no cure for silicosis, and treatment options are limited. However, the condition can be prevented if measures are taken to reduce exposure.

Symptoms

Continued exposure:

- Shortness of breath
- Fever
- Bluish skin at the ear lobes or lips

As the disease progresses:

- Fatigue
- Extreme shortness of breath
- Loss of appetite
- Chest pain
- Respiratory failure

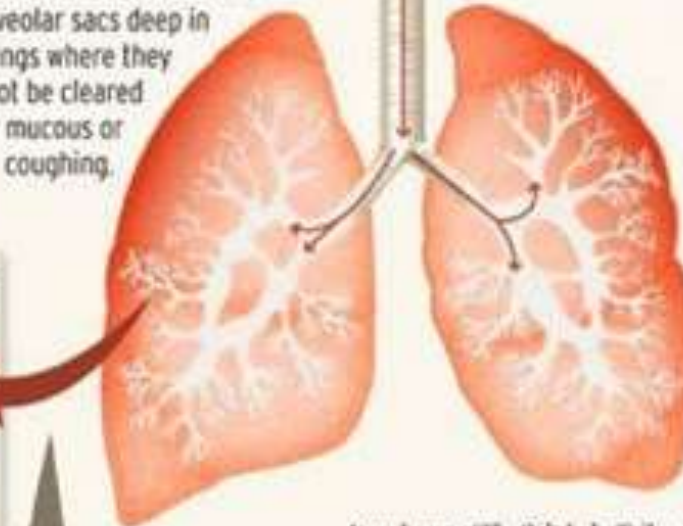
At-risk occupations

- Construction
- Mining
- Sandblasting
- Masonry
- Demolition
- Manufacturing of glass and metal products
- Plumbing
- Painting

CRYSTALLINE SILICA DUST

Inhaling the dust can cause scar tissue to form in the lungs that reduces the lungs' ability to extract oxygen from the air.

Silica dust particles can embed themselves in the alveolar sacs deep in the lungs where they cannot be cleared by mucus or coughing.



AMY LEWIS/The Salt Lake Tribune

Safety instruction!

Do wear masks while exposed to dust

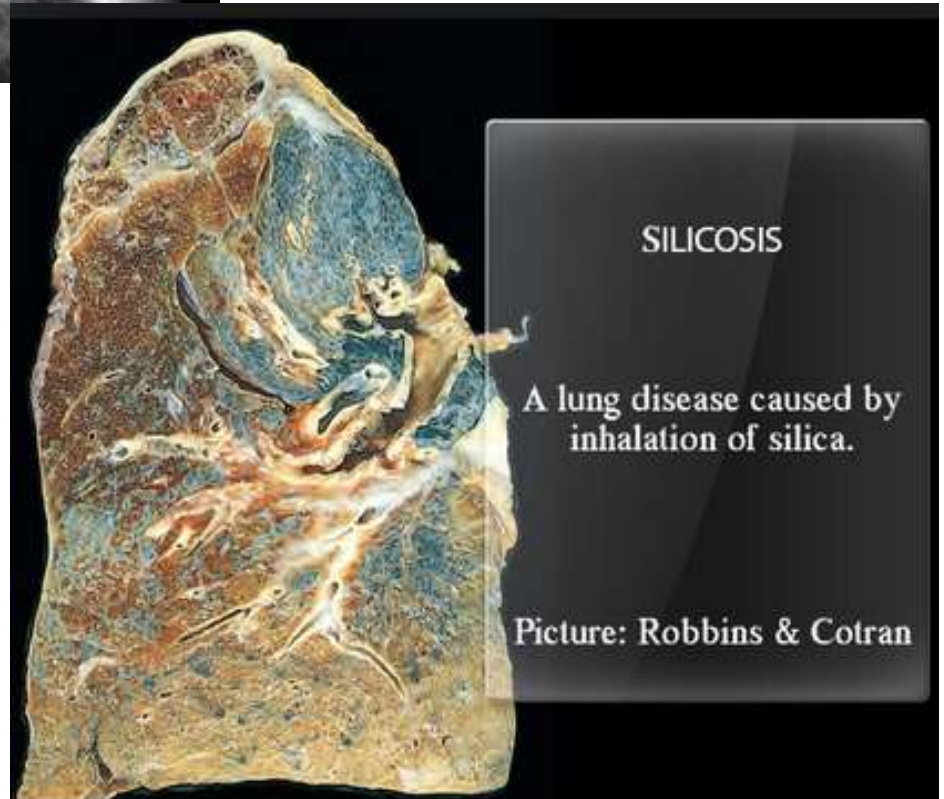


**X-ray of lung
with silicosis**

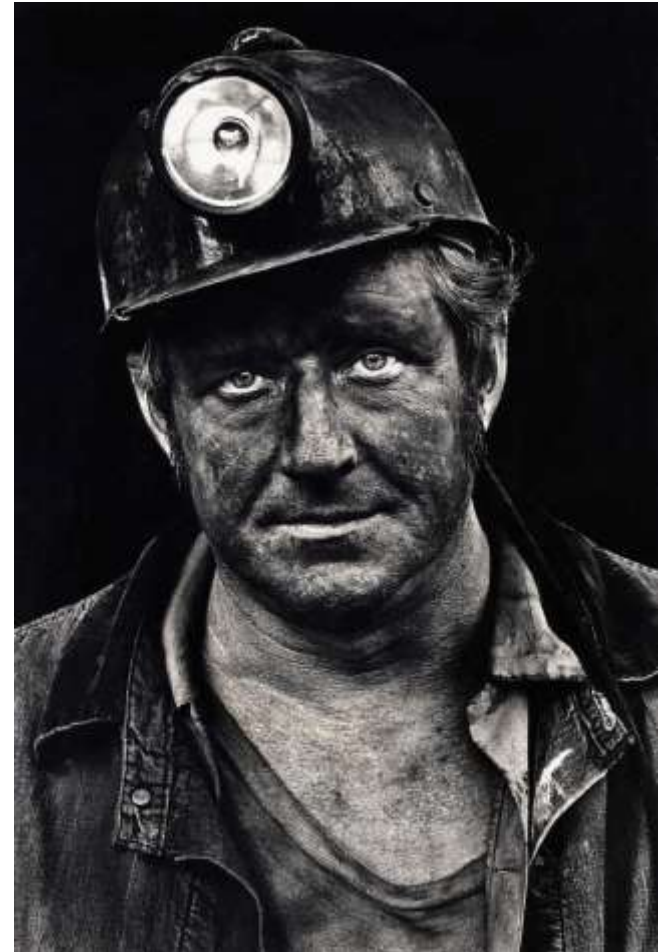
**Coal
miners
lung after
autopsy**



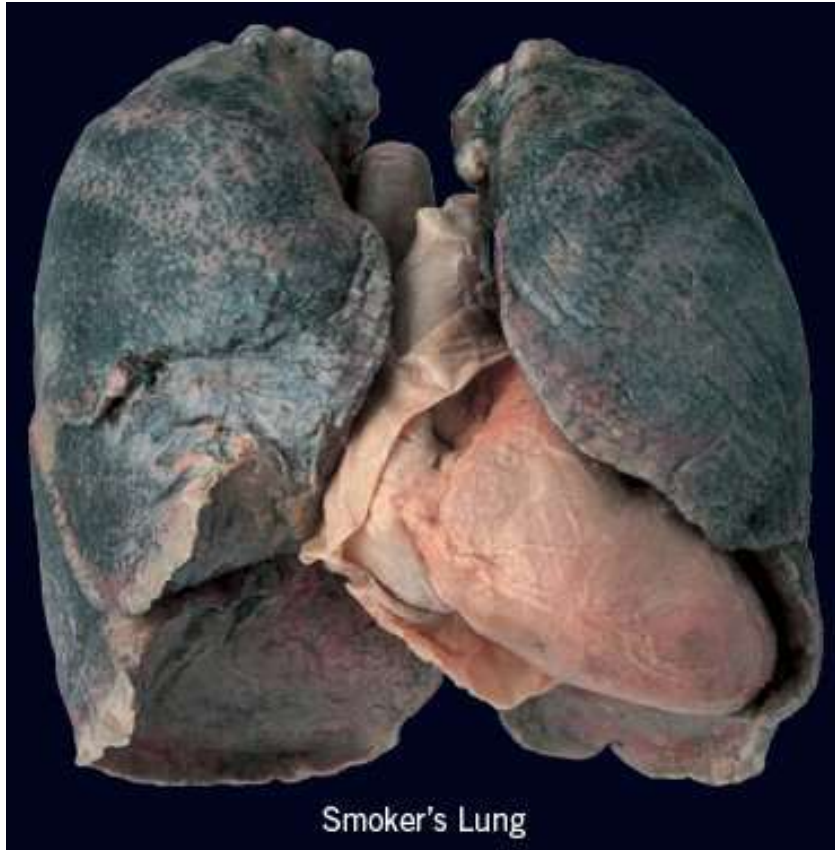
**Silicosis
lung
after
autopsy**



MINERS LUNG



SMOKERS LUNG



While BC can decompose into primary spheres and can enter the blood stream (and does), there are limits to how much BC the body processes that way.

A chain smoker (and females cooking with poor ventilation in particular when cooking using biomass burning stoves) will develop a smokers lung over time

The main difference between the lung of a coal miner and a chain smoker is that the smokers lung is blackest in the alveoli, while the miner has more dust in the bronchiolar airways